

Trader Performance vs. Bitcoin Market Sentiment

An analysis of Hyperliquid trading data against the Fear & Greed Index

Prepared for: Hiring Assignment Review

Datasets: Bitcoin Fear/Greed Index (2018–2025) · Hyperliquid Historical Trader Data (2023–2025)

Overlapping analysis window: May 2023 – May 2025 · 211,224 trade records · 32 accounts

1. Executive Summary

This report examines whether Bitcoin market sentiment — measured by the daily Fear & Greed Index — is associated with differences in how a cohort of 32 Hyperliquid accounts trade and perform. The dataset covers 211,224 trade-level records across 246 traded instruments (dominated by HYPE, BTC, ETH and SOL), matched to the daily sentiment classification by calendar date.

Four consistent patterns emerge:

- Traders are net profitable in every sentiment regime, but win rates and average profit per trade are meaningfully higher in Extreme Greed and Fear than in plain Greed or Extreme Fear — the two 'moderate' extremes outperform the milder Greed reading.
- Trading activity rises during Fear and falls during Greed: the number of trades is negatively correlated with the Fear/Greed index value ($r \approx -0.25$), meaning this cohort trades more when the market is scared.
- Positioning is contrarian at the extremes: during Extreme Fear, 68.8% of new positions opened are Longs; during Greed, only 42.3% of new positions are Longs (57.7% Shorts) — the group leans long into fear and leans short into greed.
- Average position size is largest during Fear (~\$7,375 per trade) and smallest during Extreme Greed (~\$2,780), suggesting traders commit more capital when sentiment is negative and scale down when euphoria is high.

Taken together, the data is consistent with a cohort that behaves somewhat contrarian to crowd sentiment and is rewarded for it — average PnL per trade is higher in Fear (\$112.63) than in Greed (\$85.40), and highest of all in Extreme Greed (\$130.21), which the report unpacks in Section 4.

2. Data & Methodology

2.1 Datasets

- Bitcoin Fear & Greed Index — daily classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed) with a 0–100 sentiment score, spanning Feb 2018 to May 2025.
- Hyperliquid Historical Trader Data — 211,224 individual trade executions from 32 accounts, spanning May 2023 to May 2025, with fields including account, coin, execution price, size (tokens & USD), side, direction (open/close, long/short), closed PnL, fees, and timestamp.

2.2 Approach

- Each trade's IST timestamp was normalized to a calendar date and left-joined to that day's sentiment classification (99.997% of trades matched; 6 of 211,224 fell outside the sentiment index's coverage and were excluded from sentiment-based cuts).
- Performance metrics (win rate, average/median PnL) are computed only on trades with a non-zero Closed PnL — i.e. trades that actually closed a position — since the Fee/entry legs of a position ('Open Long', 'Open Short', 'Buy', 'Sell') do not themselves realize profit or loss.
- Activity and sizing metrics (trade counts, average size) use the full trade set, since every execution — opening or closing — reflects trading behavior.
- The five-class sentiment labels were also collapsed into a simple Fear / Neutral / Greed bucket for headline comparisons, alongside the full five-class breakdown for nuance.

3. How Sentiment Shapes Trading Behavior

3.1 Trading Activity by Sentiment

Trade counts are not evenly distributed across sentiment regimes. Fear days see the most trades (61,837), and this cohort is measurably more active as sentiment sours — the correlation between the daily Fear/Greed value and daily trade count is -0.245, i.e. more trades happen on lower (fearful) sentiment days.

3.2 Position Sizing by Sentiment

Average trade size also varies by regime, peaking in Fear (~\$7,816 per trade in the plain Fear class, ~\$5,350 in Extreme Fear) and troughing in Extreme Greed (~\$3,112). This cohort appears to increase capital commitment when the market is fearful and de-risk (smaller size) when euphoria is highest — a pattern that runs opposite to the retail tendency to size up during rallies.

3.3 Long / Short Positioning by Sentiment

Looking only at newly opened positions, directional bias shifts sharply with sentiment. In Extreme Fear, 68.8% of new positions are Longs; that share falls steadily through Fear (62.1%) and Neutral (61.7%) to Greed (42.3%) and Extreme Greed (45.1%), where Shorts dominate. This is a textbook contrarian pattern — buying when others are fearful, and increasingly willing to short as the crowd turns greedy.

4. Performance by Sentiment Regime

4.1 Win Rate

Win rate (share of closing trades with positive PnL) is highest in Extreme Greed (89.2%) and Fear (87.3%), and lowest in Extreme Fear (76.2%) and plain Greed (76.9%). Neutral sits in between at 82.4%. The pattern is not a simple monotonic relationship with sentiment — both tails of pure euphoria

(Extreme Greed) and moderate fear (Fear) outperform the more middling Greed and the panic extreme of Extreme Fear.

4.2 Average Profit per Trade

Average closed PnL per trade follows a similar shape: Extreme Greed (\$130.21) and Fear (\$112.63) lead, followed by Neutral (\$71.20) and Extreme Fear (\$71.03), with plain Greed lowest (\$85.40 — still positive, but below Fear and Extreme Greed). Median PnL is positive but small across all regimes (\$4.58–\$8.53), indicating that a large share of the total profit is concentrated in a smaller number of large winning trades rather than being spread evenly.

4.3 Aggregate Performance Table

Extreme Fear	10,406	\$739,110	\$71.03	\$6.39	76.2%	\$5,468
Fear	29,808	\$3,357,155	\$112.63	\$6.35	87.3%	\$8,041
Neutral	18,159	\$1,292,921	\$71.20	\$4.58	82.4%	\$5,556
Greed	25,176	\$2,150,129	\$85.40	\$4.93	76.9%	\$5,439
Extreme Greed	20,853	\$2,715,171	\$130.21	\$8.53	89.2%	\$2,780

4.4 PnL Trend vs. Sentiment Over Time

Plotting the 7-day rolling average of daily closed PnL against the Fear/Greed index value over the full sample shows that profitable stretches are not confined to any single sentiment regime — big PnL days occur in both directions of the sentiment cycle. The overall correlation between daily PnL and the sentiment index value is weak ($r \approx -0.08$), meaning sentiment level alone is a poor direct predictor of daily profitability; the behavioral differences documented above (sizing and directional bias) are a more useful lens than the sentiment score itself.

5. Symbol-Level Patterns

The five most-traded instruments — HYPE (68,005 trades), @107 (29,992), BTC (26,064), ETH (11,158) and SOL (10,691) — show meaningfully different sentiment sensitivity:

- ETH shows its strongest average PnL during Fear (\$429.20 per trade) versus Greed (\$136.13) — a large gap suggesting ETH positions taken during fearful markets performed especially well for this cohort.
- SOL performs well broadly but is strongest in Greed (\$377.63) and Neutral (\$374.54), weaker in Fear (\$289.59) — the opposite pattern to ETH.
- @107 is the only top-5 instrument with a negative average PnL in Fear (-\$38.55), while performing well in Greed (\$238.42), indicating this token's profitability is tied more closely to risk-on conditions.

- BTC and HYPE show more modest, relatively stable average PnL across regimes (\$52–\$97 for BTC, \$35–\$72 for HYPE), suggesting these positions are less sentiment-sensitive for this cohort than the alt-coin exposures.

This suggests sentiment-aware strategy design should be instrument-specific rather than applied as a single blanket rule across a portfolio.

6. Account-Level Observations

29 of the 32 accounts in the dataset are net profitable over the full sample, indicating this is a skewed, largely experienced cohort rather than a representative retail sample — a caveat worth keeping in mind when generalizing these findings to the broader market.

Because performance and behavior both vary by account, sentiment-based strategy recommendations below are framed at the cohort level; a natural next step (noted in Section 8) would be to test whether the contrarian sizing/direction pattern holds within each individual account or is driven by a subset of large traders.

7. Implications for Trading Strategy

- Position sizing: The data supports scaling size up moderately during Fear/Extreme Fear and down during Extreme Greed — consistent with what this profitable cohort already does, and consistent with buying volatility/dislocation rather than chasing momentum.
- Directional bias: A sentiment-aware overlay that tilts toward Long entries during Fear/Extreme Fear and toward Short entries during Greed/Extreme Greed aligns with the cohort's revealed behavior and its favorable win rates in Fear and Extreme Greed.
- Instrument selection: Sentiment sensitivity is coin-specific (e.g., ETH performing best in Fear, SOL and @107 performing best in Greed). A single sentiment rule should not be applied uniformly across a multi-asset book; per-instrument backtests are warranted before allocating a strategy this way.
- Activity throttling: Since trade frequency already rises in Fear for this cohort, a systematic strategy could formalize this — increasing scan/execution frequency specifically when the index is in Fear/Extreme Fear territory.
- Caution on plain 'Greed': Both tails (Extreme Fear and Extreme Greed) plus plain Fear show stronger metrics than plain Greed in this data. Greed being the weakest win-rate regime (76.9%) is a useful, slightly counter-intuitive flag: it may be the point where the crowd trade is 'obvious' but not yet euphoric enough to be reliably profitable — worth further investigation before acting on it.

8. Limitations & Suggested Next Steps

- The sample covers 32 accounts already selected for inclusion in a Hyperliquid dataset — results may not generalize to the broader trading population and are more reflective of this specific, largely profitable cohort.
- Sentiment is matched at daily granularity; intraday sentiment shifts (e.g., a Fear day that turns to Greed within hours) are not captured.
- Correlation between sentiment score and PnL is weak in aggregate (Section 4.4); the more informative signal in this data is behavioral (sizing, direction) rather than the sentiment score predicting PnL directly — this distinction matters for how any strategy would be built and

validated.

- Recommended next steps: (1) test a lagged sentiment window (does yesterday's sentiment predict today's performance better than same-day?); (2) run instrument-specific backtests of the contrarian-sizing rule; (3) check whether findings hold when accounts are analyzed individually rather than pooled, to rule out a few large traders driving the aggregate pattern.

— *End of Report* —